

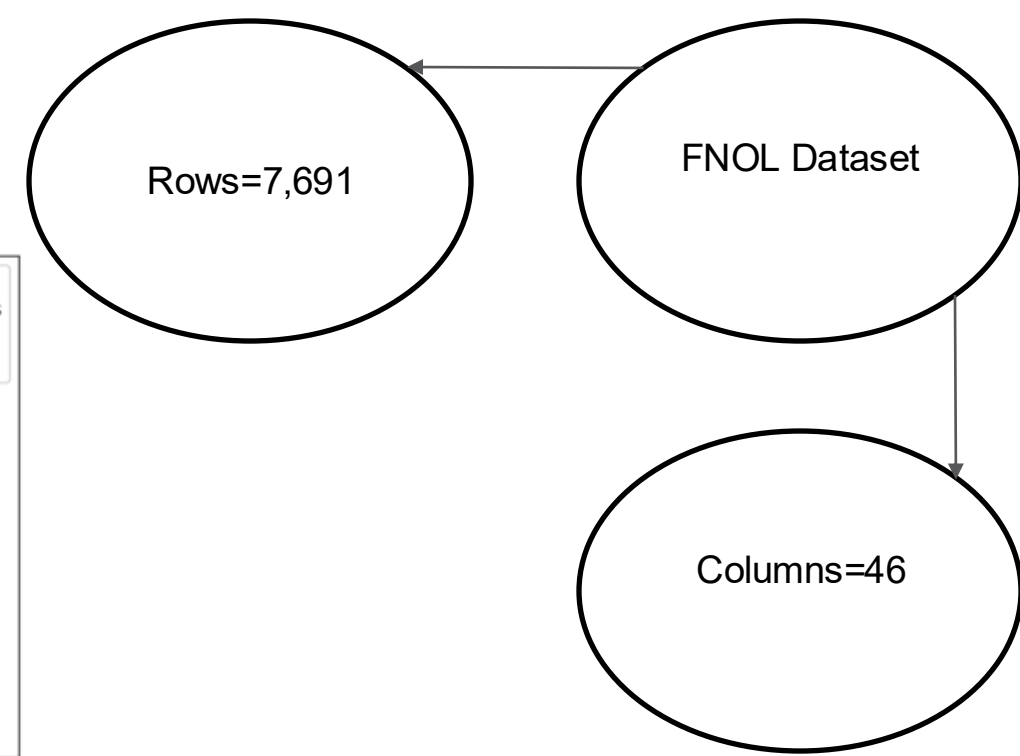
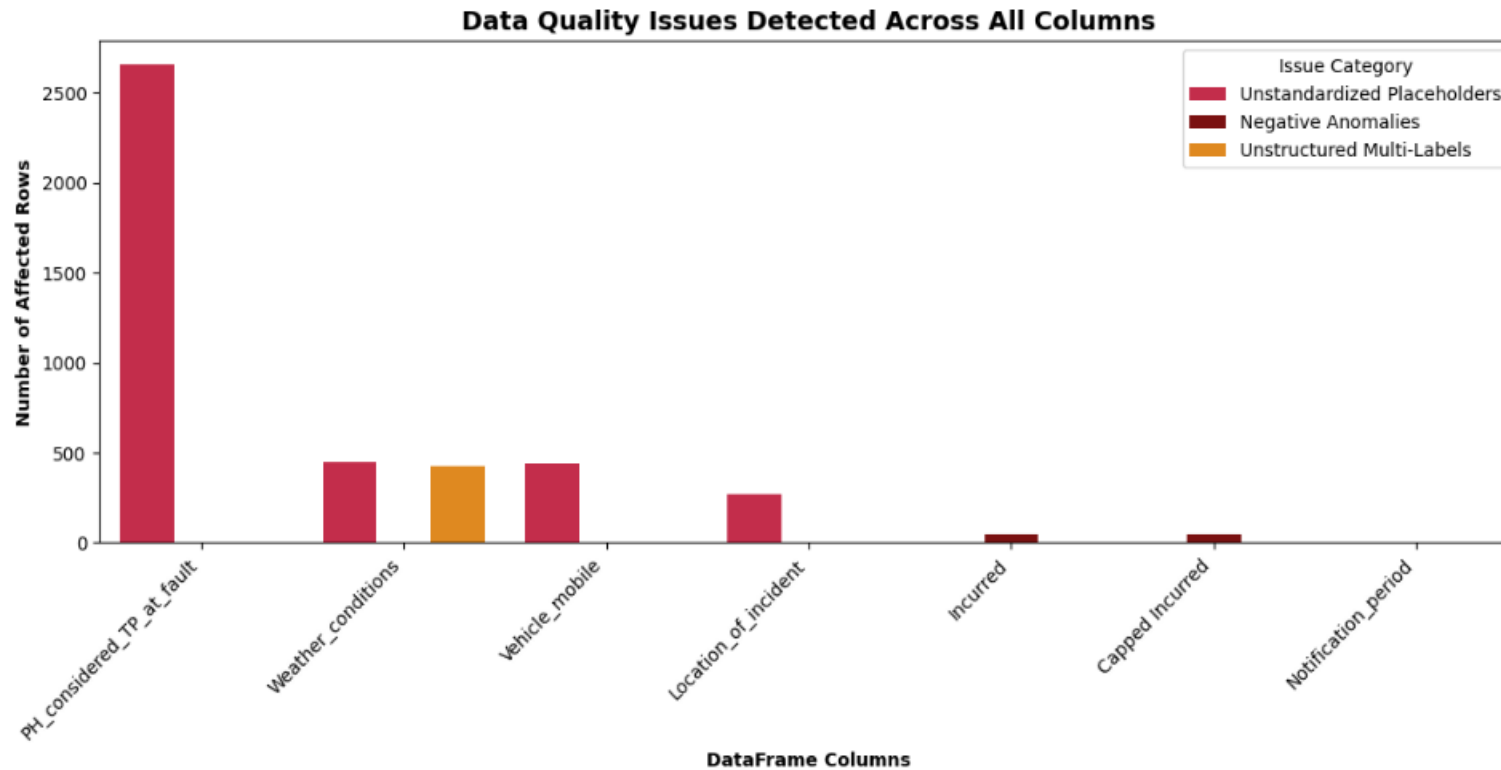
APP - Senior Data Scientist Case Study

Explore whether the ultimate cost can be predicted meaningfully from first notification of loss (FNOL) characteristics.



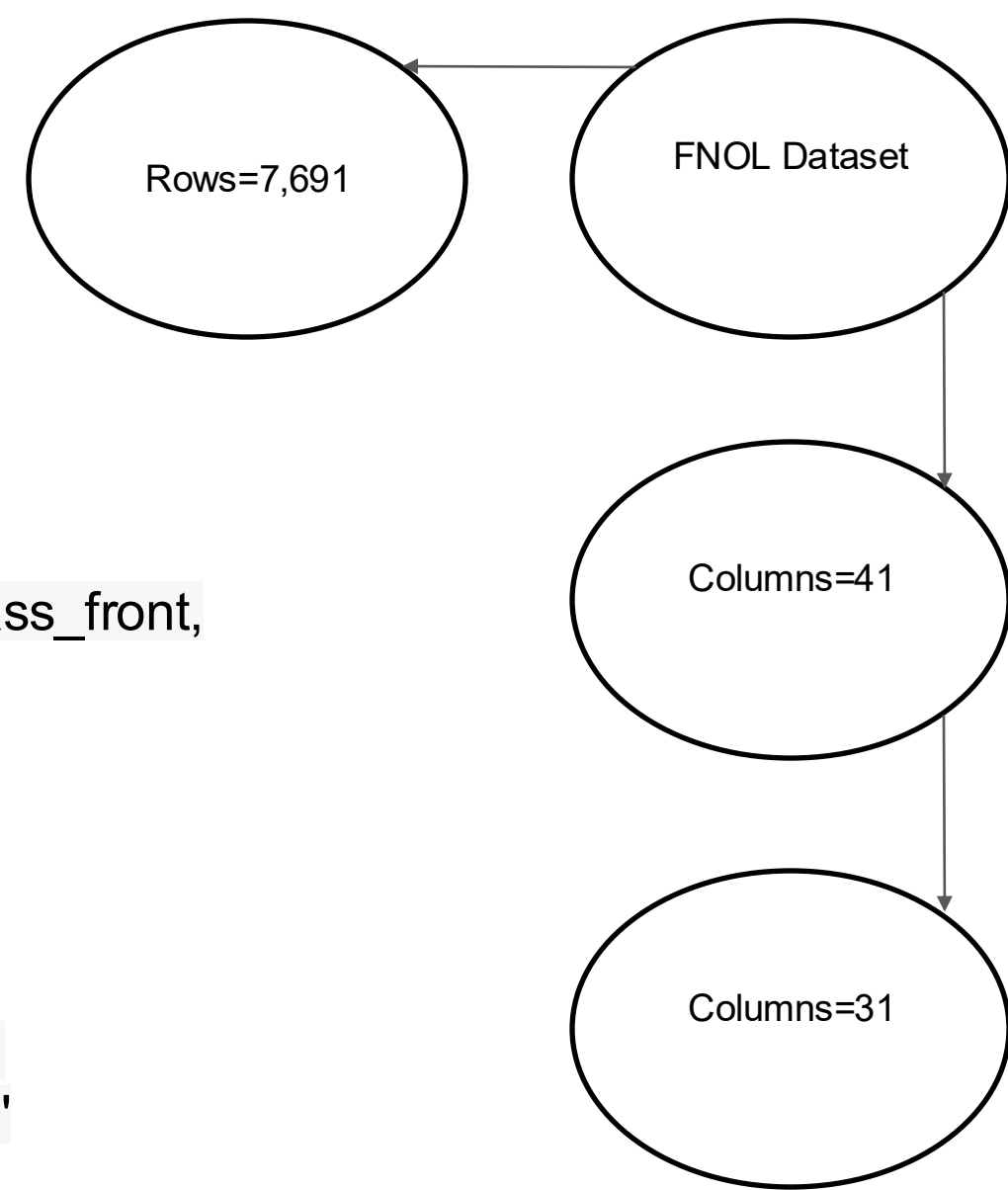
- Data Cleansing & Features engineering
- Model Choice
- Model Evaluation
- Improvements
- Practical Considerations

FNOL Data Quality Issues



- 1) **Unstandardized missing values like: 'n/k', 'N/K', '-', '#', 'Not Applicable'**
Standardized all missing placeholders to true NaNs
- 2) **Anomalies / Negative Values: Notification_period or Capped Incurred can not be negative**
Remove invalid or negative values records
- 3) **Unstructured Categories like 'SNOW, ICE, FOG' in weather condition**
Leave it as it is

Data Cleansing & Features engineering



Drop 5 Zero variance & Unhelpful features:

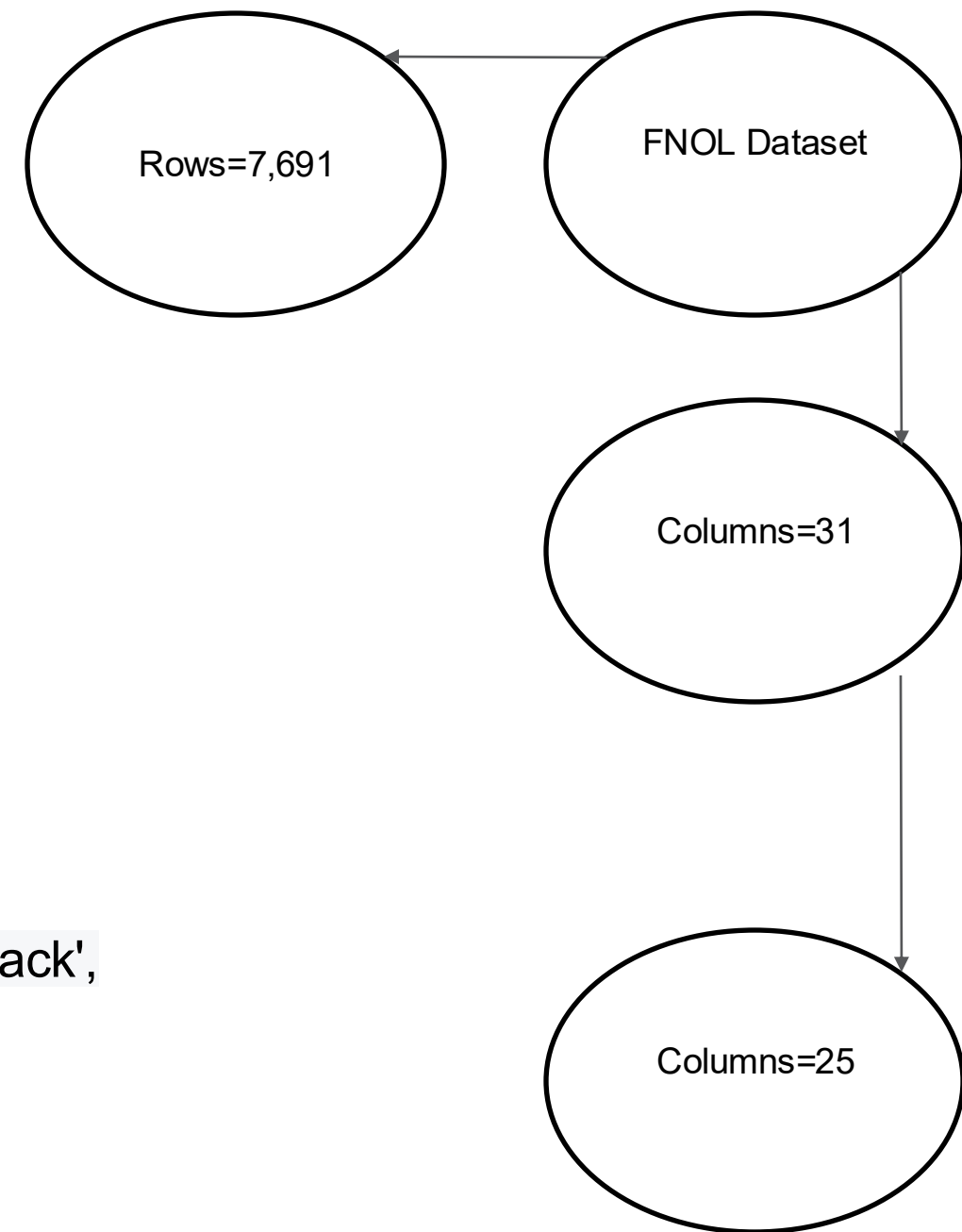
Claim Number, Loss_code, Loss_description, TP_type_insd_pass_front, TP_type_pass_multi

Features engineering (Regions):

'TP_region_eastang', 'TP_region_eastmid', 'TP_region_london', 'TP_region_north', 'TP_region_northw', 'TP_region_outerldn', 'TP_region_scotland', 'TP_region_southe', 'TP_region_southw', 'TP_region_wales', 'TP_region_westmid', 'TP_region_yorkshire'

Transformed 12 to 2 columns (Total_Regional_TP_Count, Primary_TP_Region)

Data Cleansing & Features engineering



Features engineering (Third party type):

'TP_type_insd_pass_back', 'TP_type_driver', 'TP_type_pass_back',
'TP_type_pass_front', 'TP_type_bike', 'TP_type_cyclist',
'TP_type_pedestrian', 'TP_type_other', 'TP_type_nk'

Transformed 9 to 3 columns (Total_TP_Passenger_Count,
Has_Vulnerable_Road_User, Total_TP_Type_Count)

Data Cleansing & Features engineering

Features engineering (Third party injury):

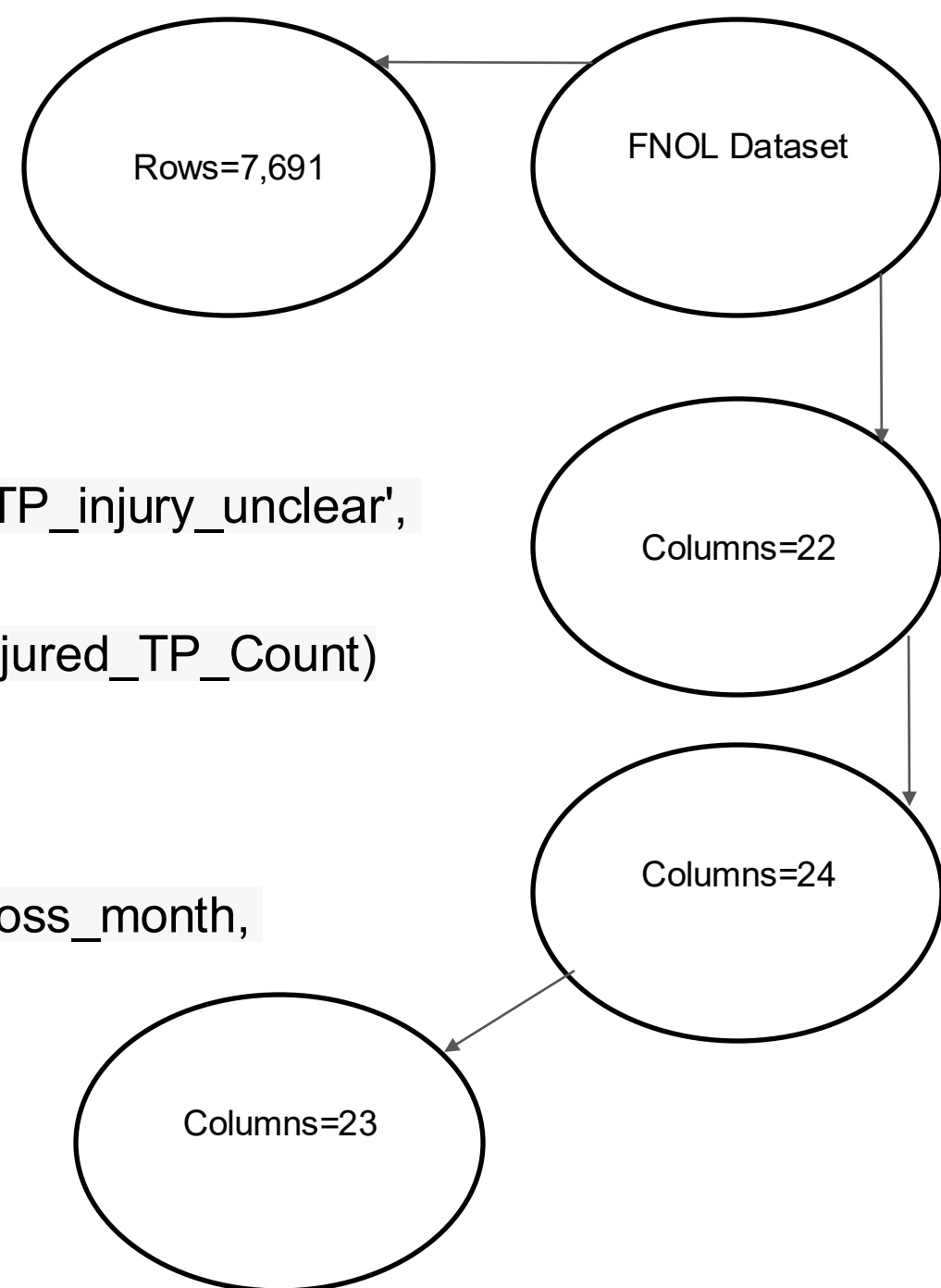
'TP_injury_whiplash', 'TP_injury_traumatic', 'TP_injury_fatality', 'TP_injury_unclear', 'TP_injury_nk'

Transformed 5 to 2 columns (Max_Injury_Severity_Tier, Total_Injured_TP_Count)

Features engineering(date_of_loss) :

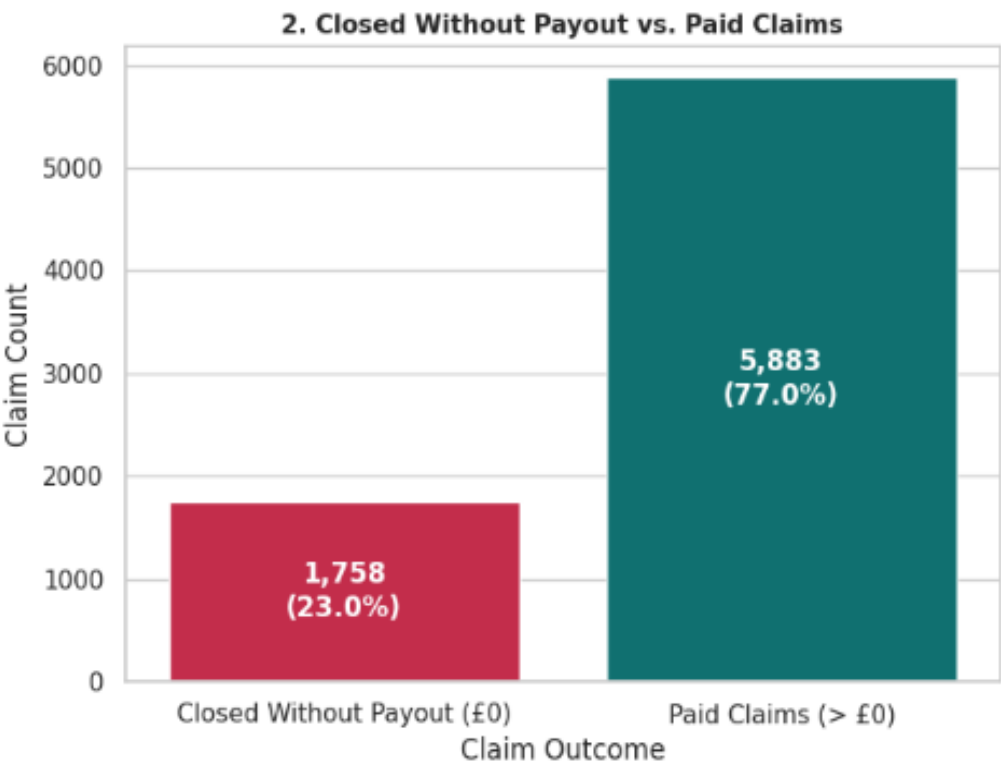
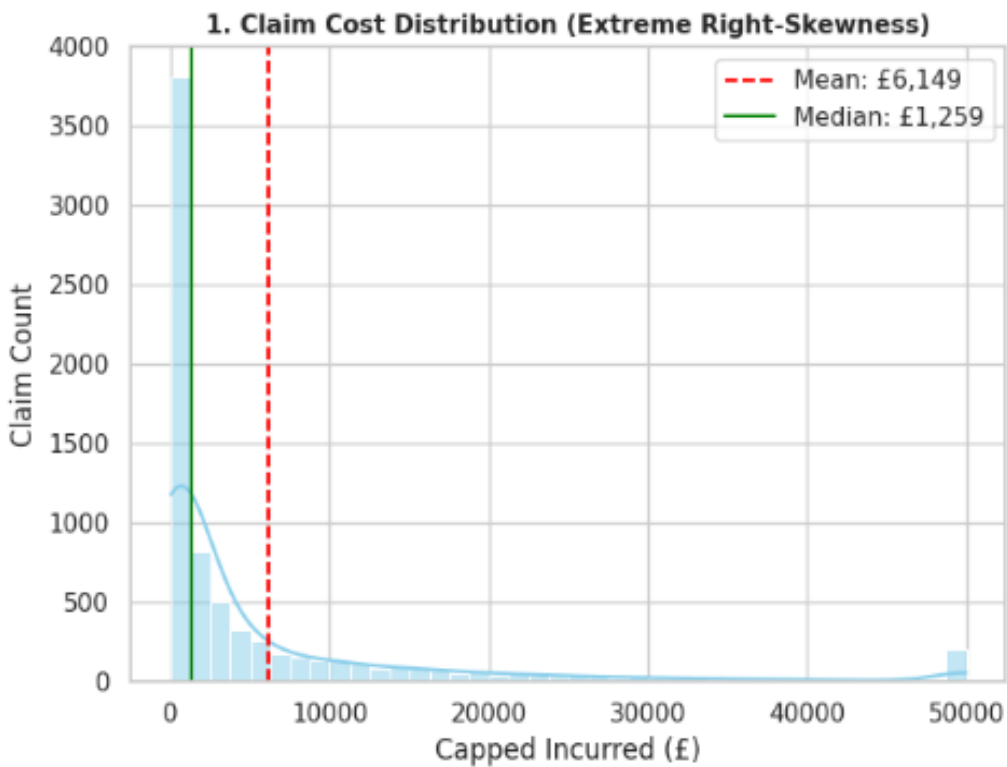
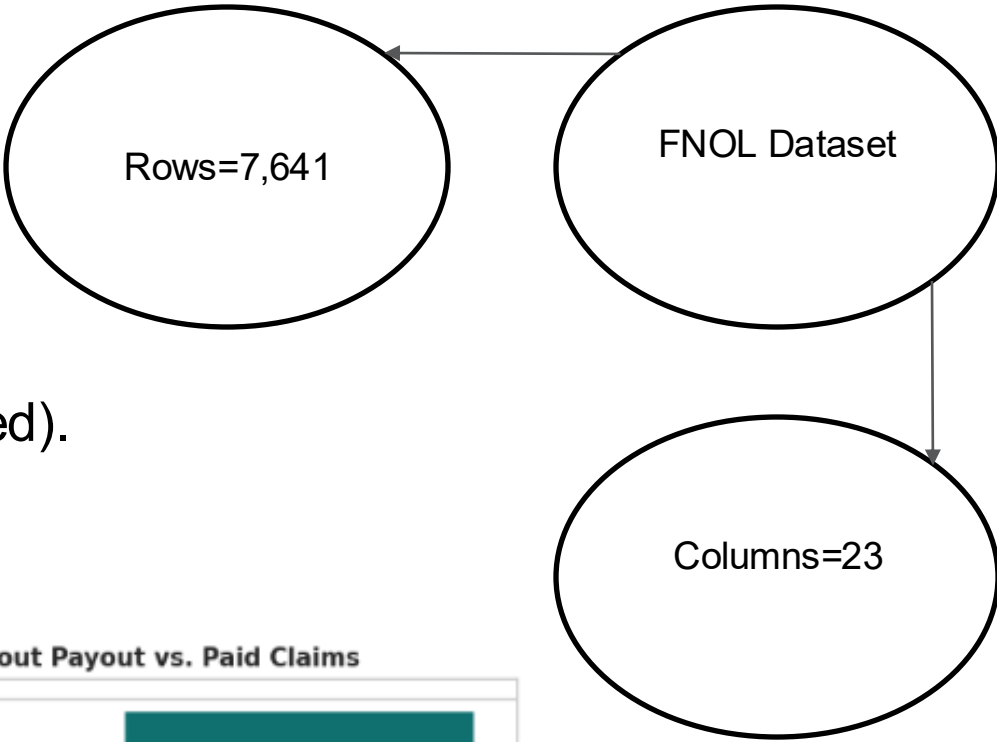
Transformed data_of_loss column to three columns (loss_year, loss_month, loss_day_of_week)

Drop Incurred column



Target

Aim : Regression task predicting claim severity (Capped Incurred).



Model Choice

There are two approaches:

- 1) Two stages modeling:
 - Use binary classification model on all data to predict probability of payout
 - Use regression model on only paid claim records to predict claim's cost
- 2) One stage modeling:
 - Use regression model on all records to predict claim's cost

Classification Models:

- 1) Logistic Regression
- 2) Decision Tree
- 3) Random Forest
- 4) Gradient Boosting Models



Regression models:

- 1) Linear Regression
- 2) Decision Tree
- 3) Random Forest
- 4) Gradient Boosting Models

Model Choice

1) Logistic Regression and linear regression

I didn't choose linear or logistic regression because extreme right-skewness creates high-leverage outliers that distort model coefficients, requiring functions like log or Box-Cox transformations to squash long tails into a normal bell curve. whereas XGBoost handles skewness automatically.

2) Decision Tree

Because of lack of regularization penalties(L1 and L2) High Risk of Overfitting.

3) Random Forest

Bootstrap Aggregating can control over fitting but it is not perfectly control the cost function

4) Gradient Boosting Models

Gradient Boosting builds trees **sequentially**. Each new tree learns specifically from the errors of the previous trees. This enables the model to capture subtle, complex risk signals that other algorithms miss and mathematically suitable for right skewed data.

Automatically handle missing value and categorical features.



Classification Metrics



True Positive (TP): - Reality: Malignant - ML model predicted: Malignant - Number of TP results: 1	False Positive (FP): - Reality: Benign - ML model predicted: Malignant - Number of FP results: 1
False Negative (FN): - Reality: Malignant - ML model predicted: Benign - Number of FN results: 8	True Negative (TN): - Reality: Benign - ML model predicted: Benign - Number of TN results: 90

Accuracy=(TP+TN)/(TP+TN+FP+FN)=(1+90)/(1+90+1+8)=0.91 (number of correct predictions)

Precision=TP/(TP+FP)=1/(1+1)=0.5 (measures how well a model avoids **False Positives**)

Recall=TP/(TP+FN)=1/(1+8)=0.11 (measures how well a model avoids **False Negatives**)

F1 score = $2 \times (\text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})) = 2 \times (0.5 \times 0.11 / (0.5 + 0.11)) = 0.18$ (**harmonic mean**)

AUC-ROC = Calculate FPR & TPR in each prediction and make a graph and calculate area under curve.

AUC-PR = Calculate precision & recall in each prediction and make a graph and calculate area under curve.

Log Loss = measures how well a classification model's probabilistic outputs align with the actual true labels.

Two Stages Modeling

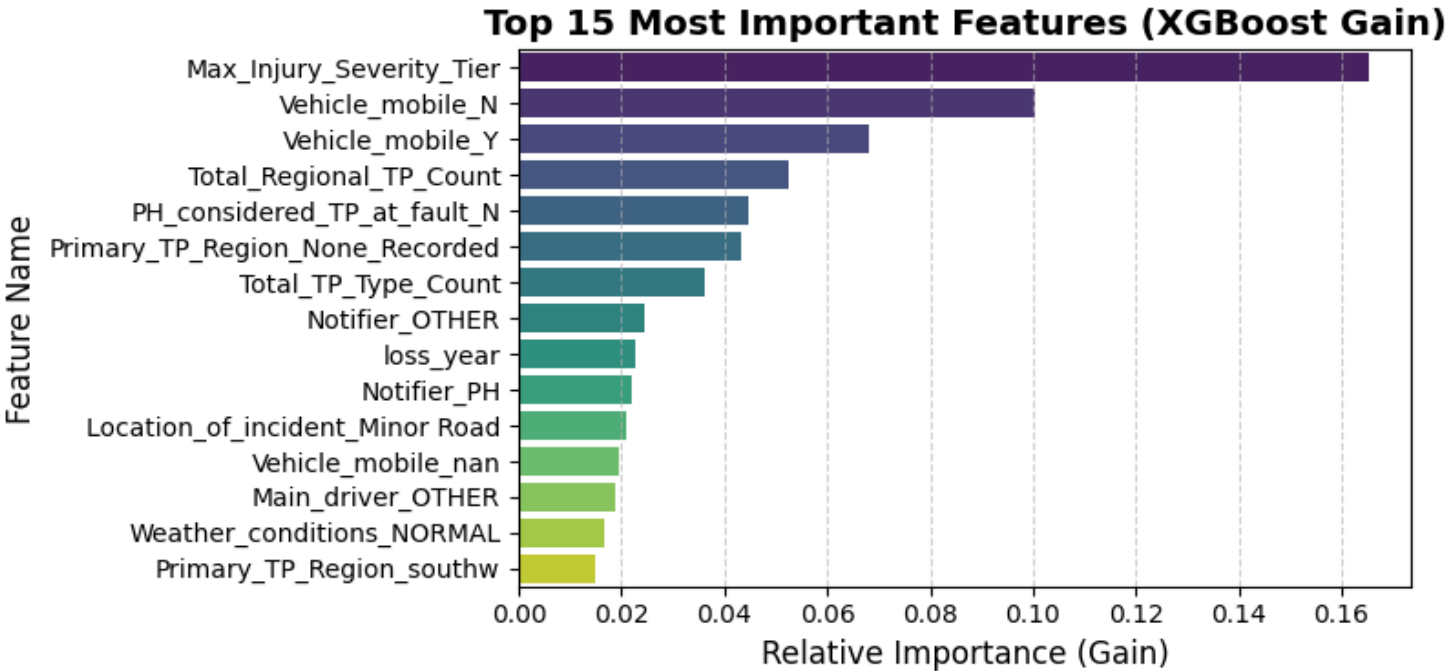
Binary classification model on all data to predict probability of payout

The XGBoost binary classification best model

F1: 0.8643 | Precision: 0.7769 | Recall: 0.9737 | PR-AUC: 0.9078 | Log Loss: 0.4703

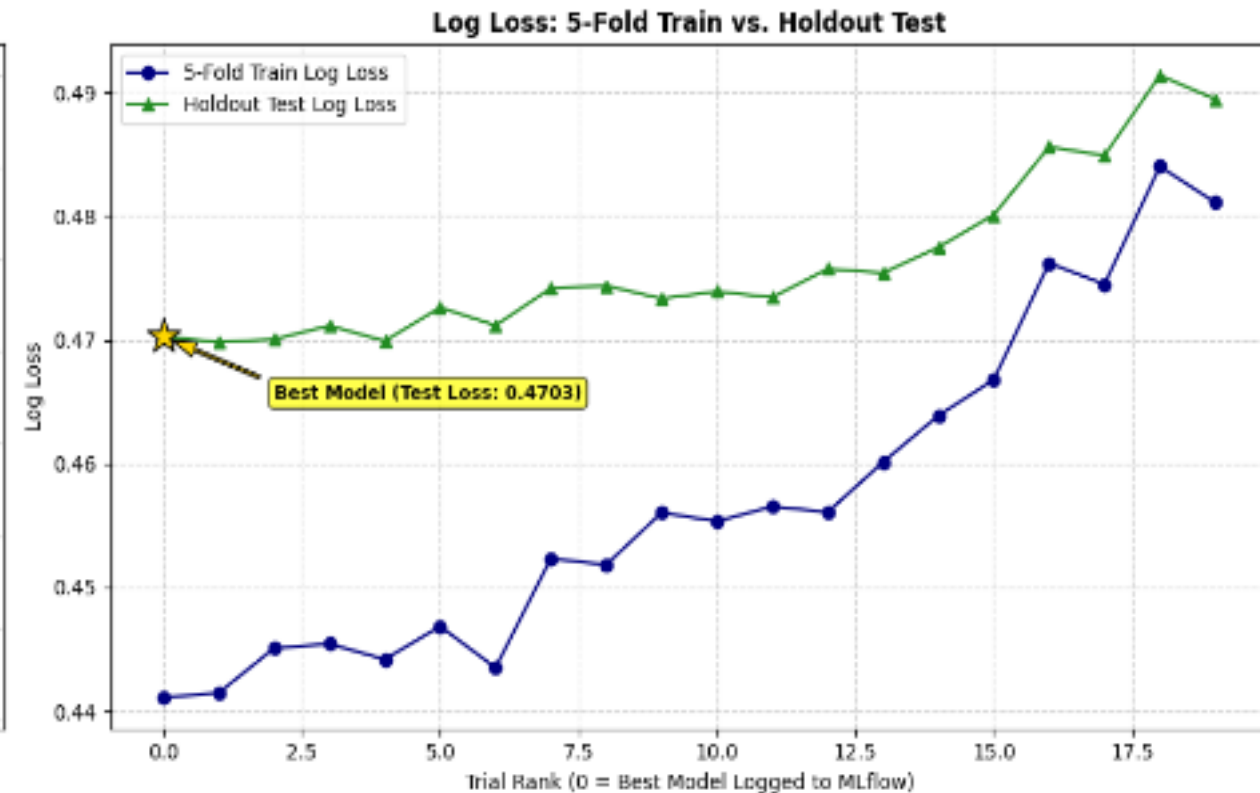
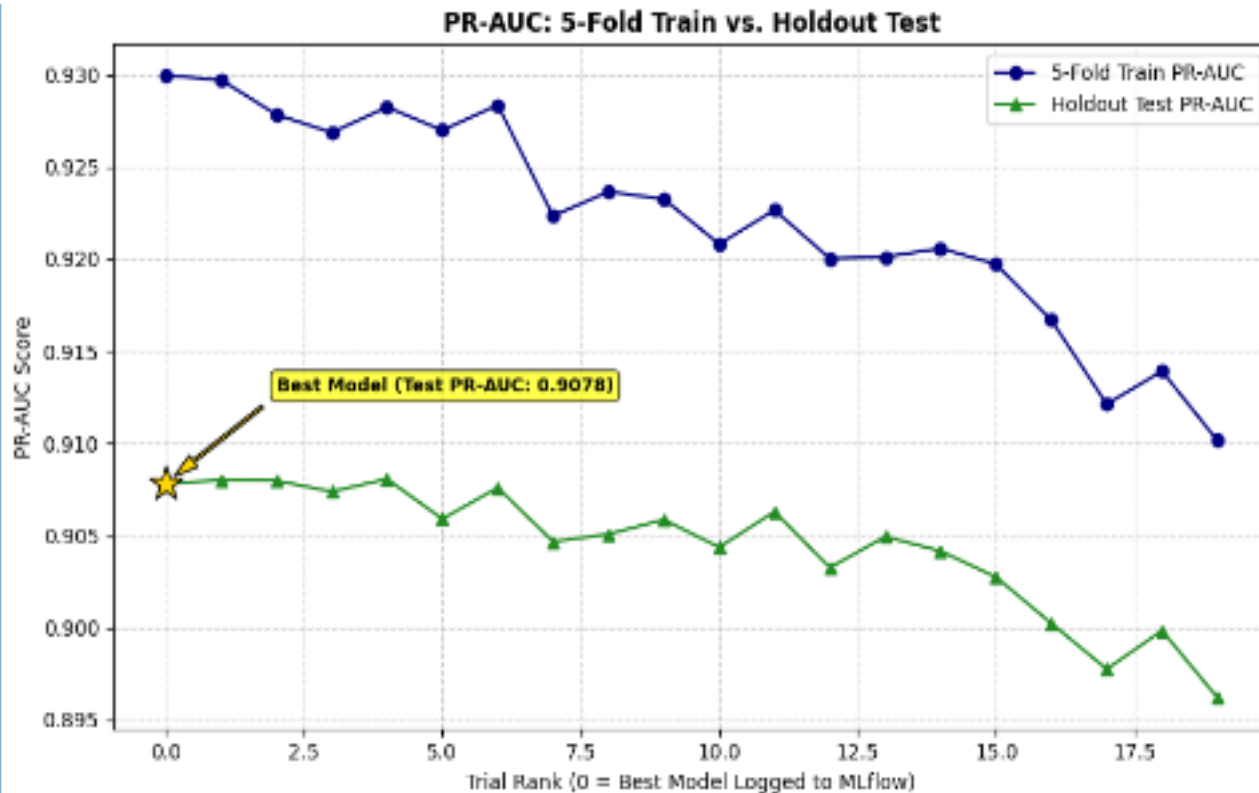
Detailed Classification Report:

	precision	recall	f1-score	support
No Payout (0)	0.43	0.07	0.11	352
Payout (1)	0.78	0.97	0.86	1177
accuracy			0.76	1529
macro avg	0.60	0.52	0.49	1529
weighted avg	0.70	0.76	0.69	1529



Two Stages Modeling

The XGBoost binary classification best model is not overfit.



Two Stages Modeling

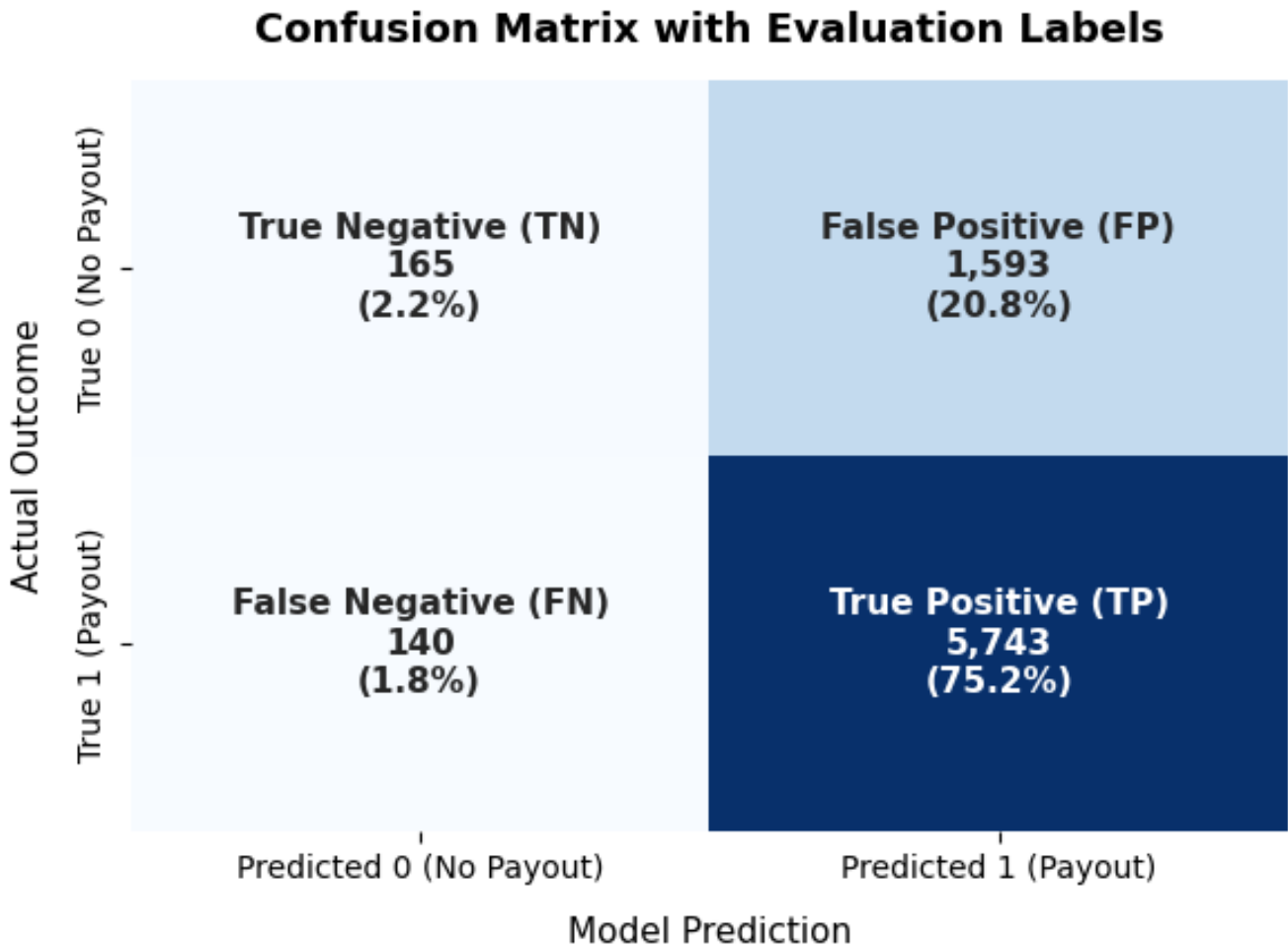
The XGBoost binary classification best model Performance on FNOL dataset.

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FULL DATASET PERFORMANCE EVALUATION

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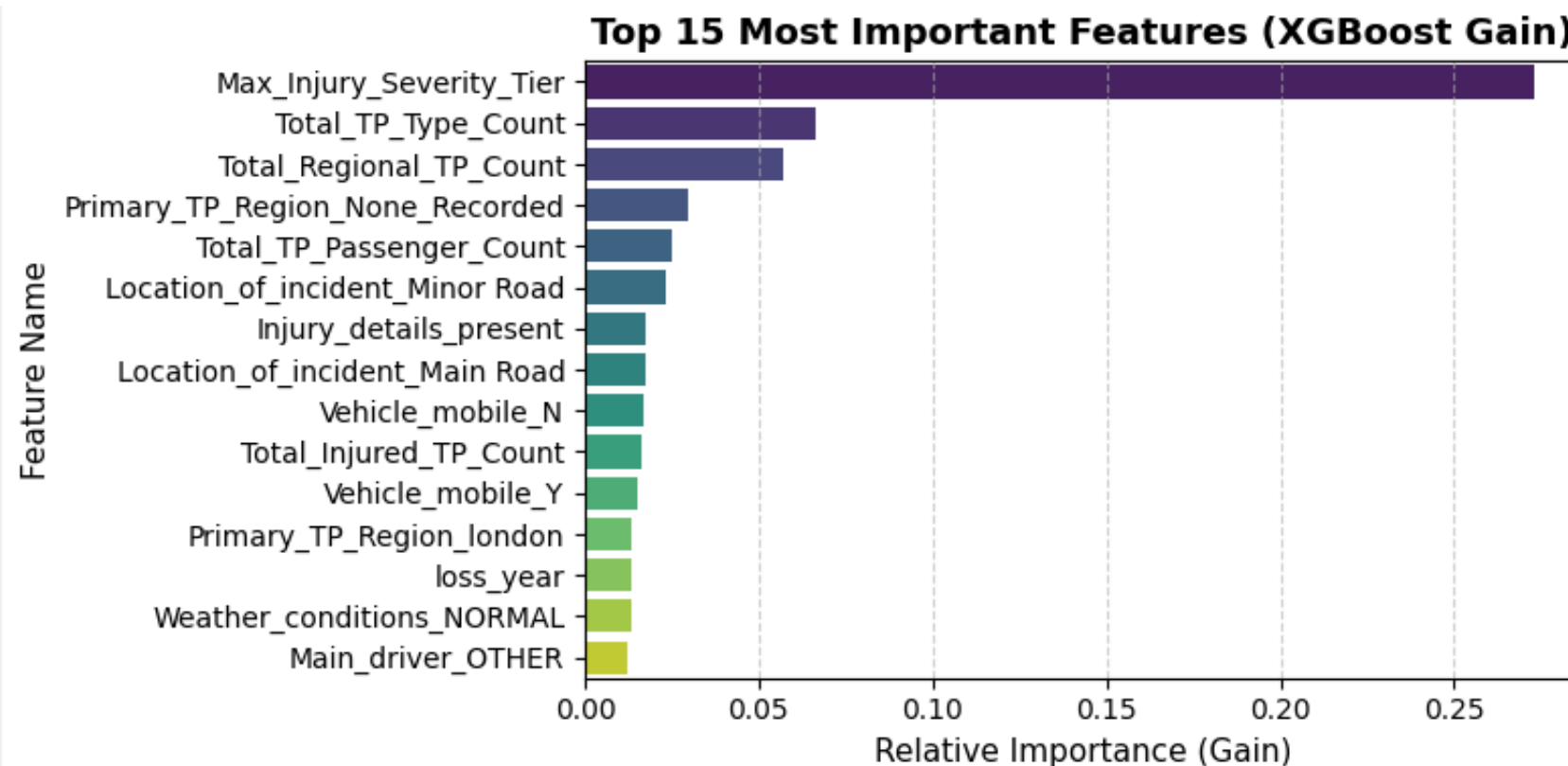
Total Claims Analyzed: 7,641				
Overall F1-Score: 0.8689				
Overall PR-AUC: 0.9255				
Overall Log Loss: 0.4465				
	precision	recall	f1-score	support
No Payout (0)	0.54	0.09	0.16	1758
Payout (1)	0.78	0.98	0.87	5883
accuracy			0.77	7641
macro avg	0.66	0.54	0.51	7641
weighted avg	0.73	0.77	0.71	7641



Two Stages Modeling

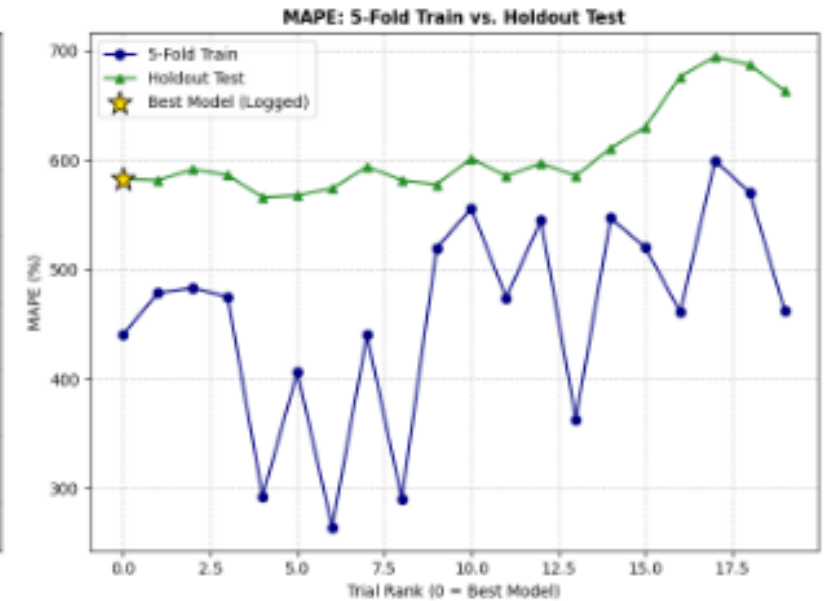
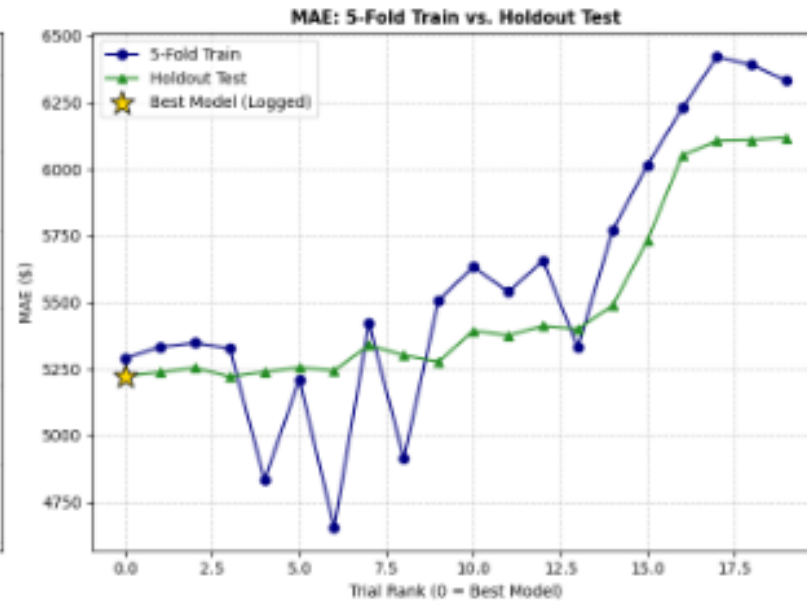
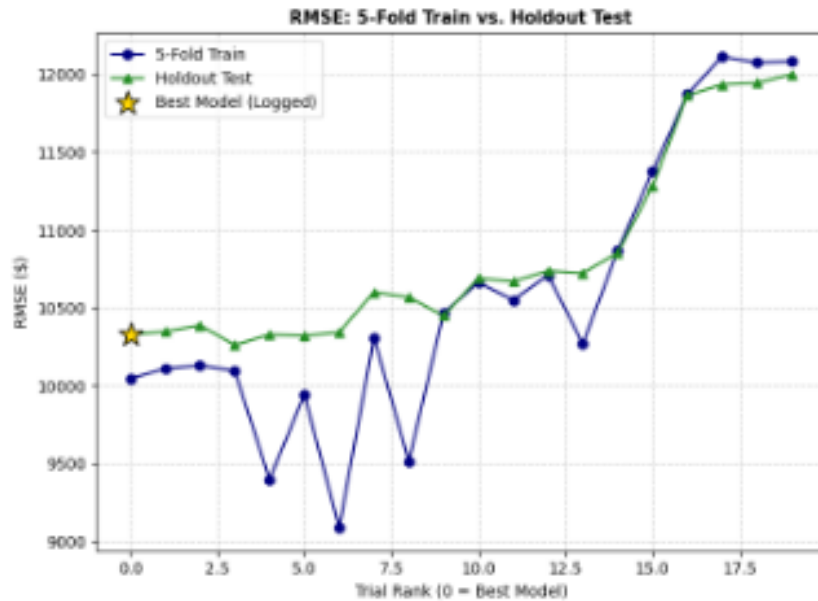
Regression model on only paid claim records to predict claim's cost
The XGBoost Regression best Model

RMSE: \$10329.16 | MAE: \$5222.63 | MAPE: 582.37% | R^2 Score: 0.2531



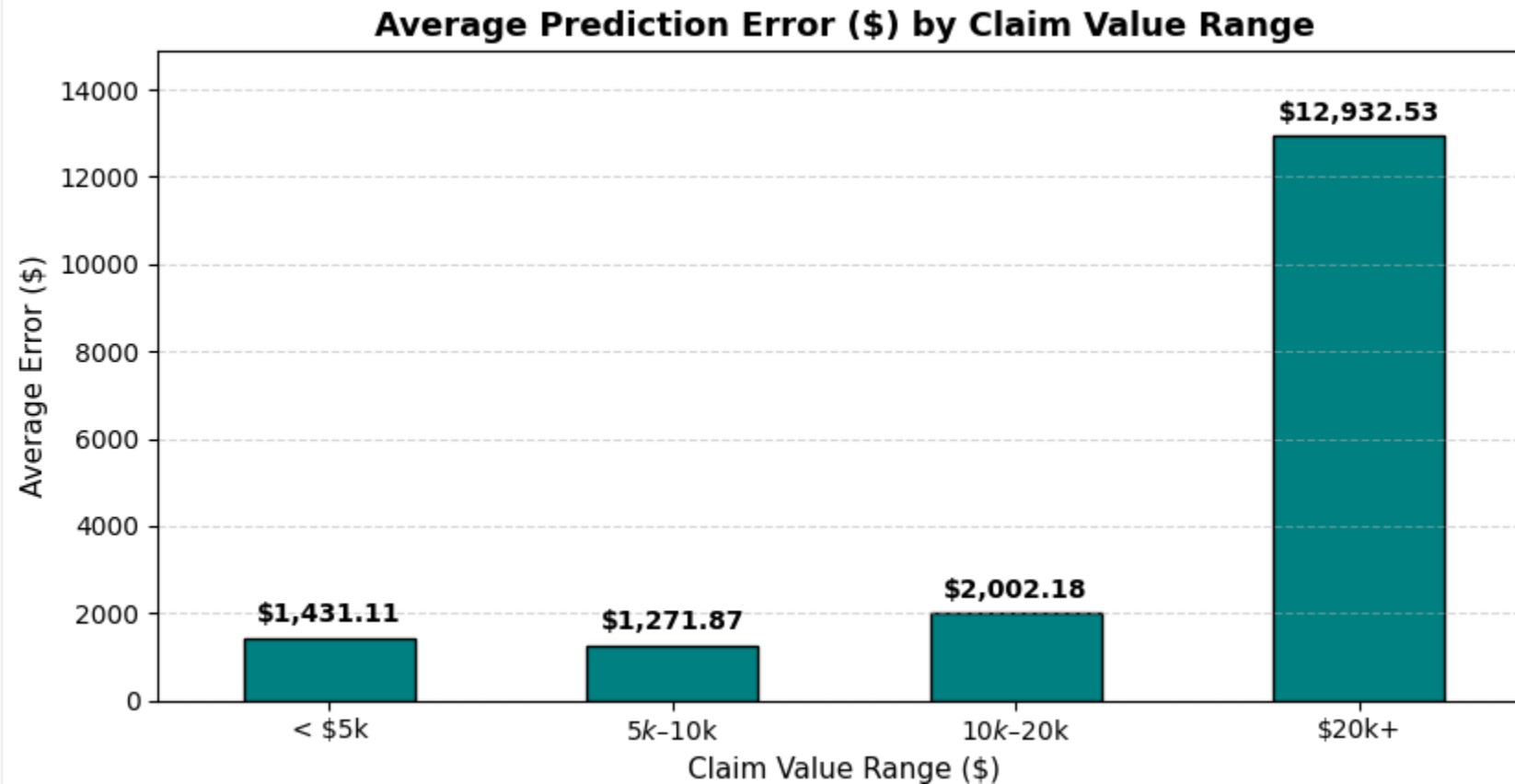
Two Stages Modeling

The XGBoost Regression best Model is not overfit.



Two Stages Modeling

The XGBoost Regression best Model Performance on FNOL dataset.



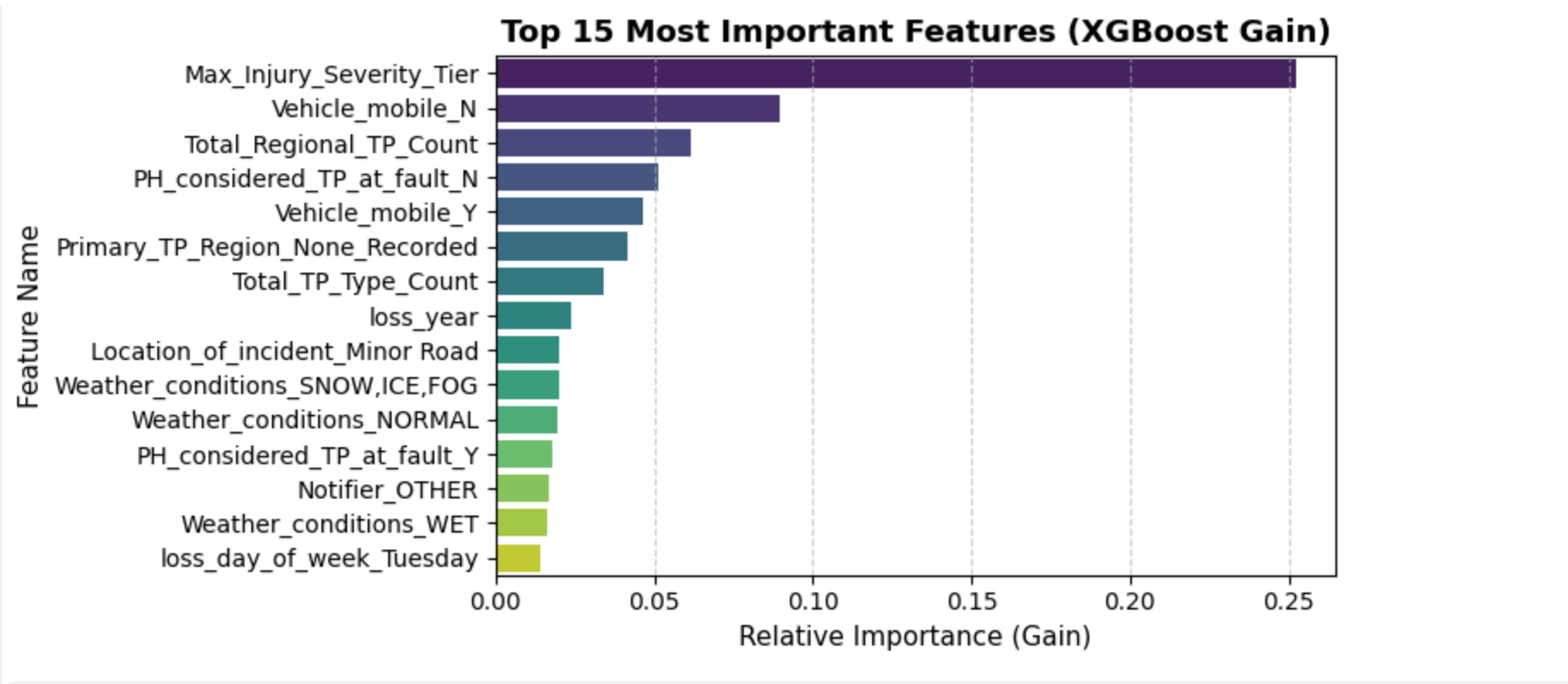
	Claim Value Range	Claim Count	Percentage (%)
0	< \$5k	5465	71.52
1	\$5k-\$10k	722	9.45
2	\$10k-\$20k	715	9.36
3	\$20k+	739	9.67

One Stage Modeling

Regression model on all records to predict claim's cost

The XGBoost Regression best Model

RMSE: \$9515.90 | MAE: \$4463.49 | MAPE: 660767409.72% | R² Score: 0.1797



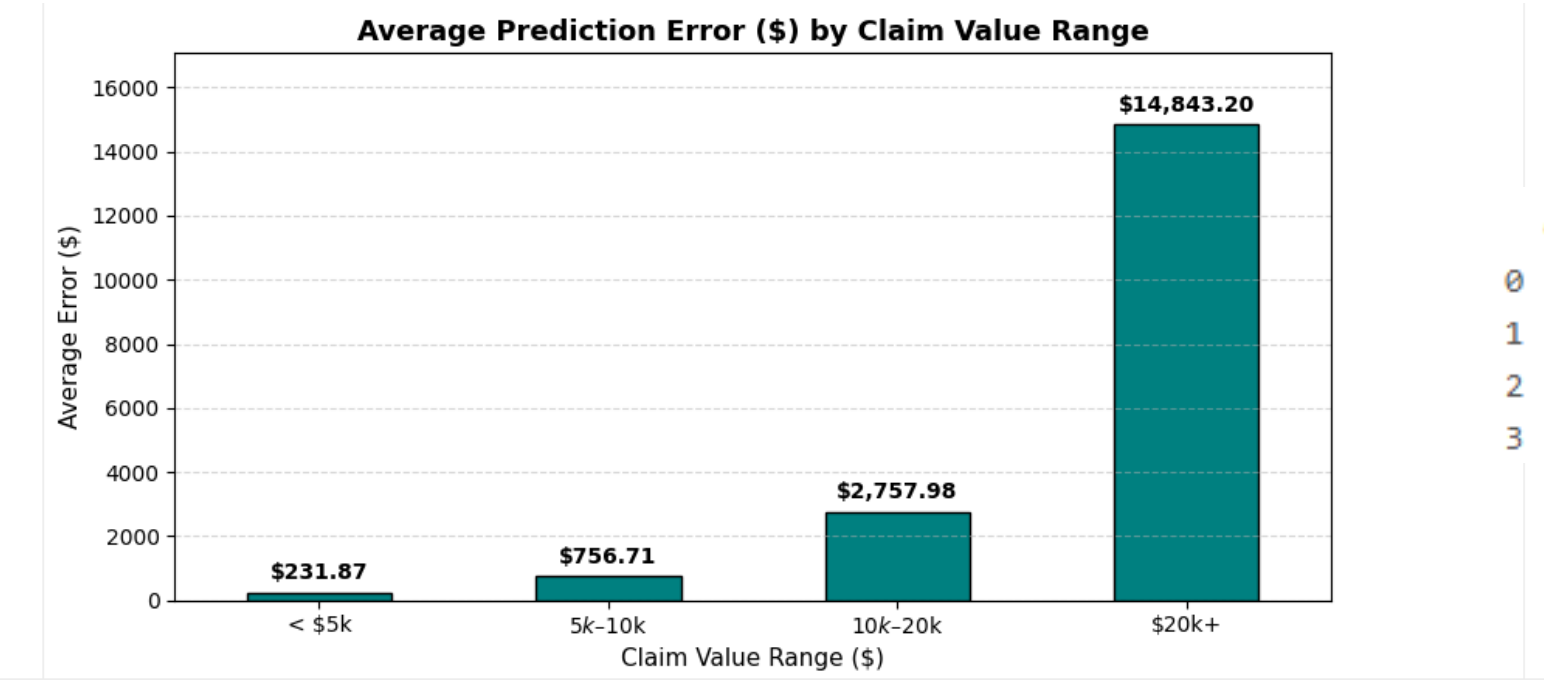
One Stage Modeling

The XGBoost Regression best Model is not overfit.



One Stage Modeling

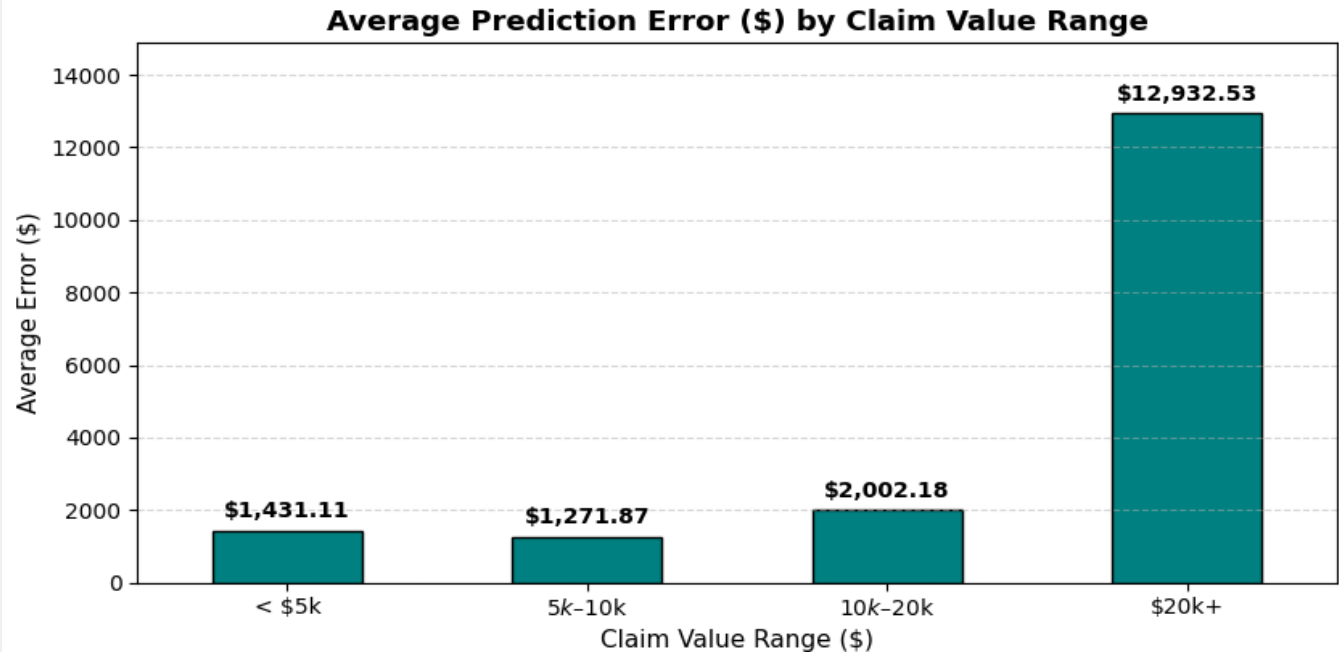
The XGBoost Regression best Model Performance the model on FNOL dataset.



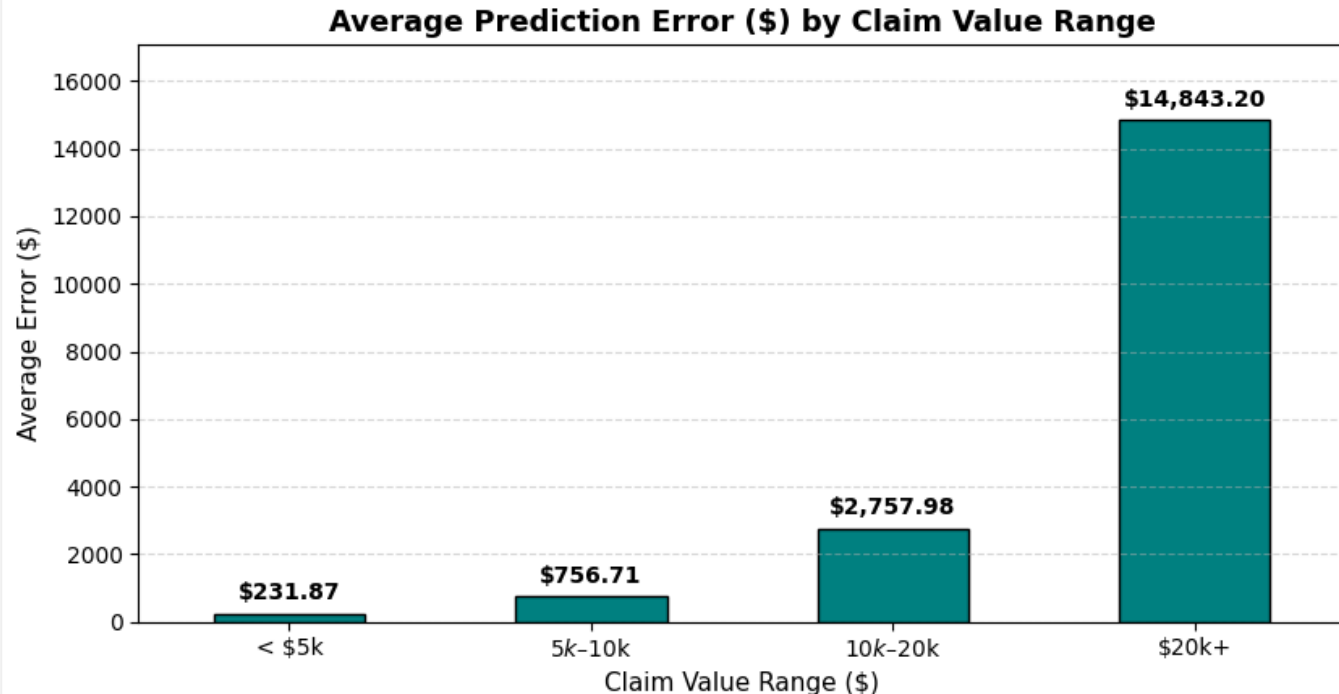
	Claim Value Range	Claim Count	Percentage (%)
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Compare Two approaches

Regression model
on only payout
claim records



Regression model
on all records



Practical Considerations:



We should index historical claims to 2026 RPI values pre-training so the model predicts actual claim severity in today's pounds.